

0620 • PAPER 4

TOPIC

06

Chemical reactions

Questions only • Original examination pages retained

11

QUESTIONS

132

TOTAL MARKS

1

SECTIONS



USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

Quick navigation and progress overview

1/1

01

Rates, equilibrium, redox and displacement

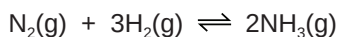
11 questions • 132 marks

6.1 Rates, equilibrium, redox and displacement

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	2021 Feb/March Paper 42 Question 3	12	☐	☐	☐
02	2021 May/June Paper 41 Question 4	7	☐	☐	☐
03	2021 May/June Paper 43 Question 4	7	☐	☐	☐
04	2021 Oct/Nov Paper 43 Question 6	10	☐	☐	☐
05	2022 Oct/Nov Paper 41 Question 3	14	☐	☐	☐
06	2023 May/June Paper 41 Question 4	14	☐	☐	☐
07	2023 May/June Paper 42 Question 3	17	☐	☐	☐
08	2023 Oct/Nov Paper 41 Question 5	12	☐	☐	☐
09	2024 May/June Paper 41 Question 4	10	☐	☐	☐
10	2024 May/June Paper 43 Question 4	10	☐	☐	☐
11	2024 Oct/Nov Paper 42 Question 3	19	☐	☐	☐

3 This question is about ammonia.

(a) Nitrogen reacts with hydrogen to form ammonia in an industrial process.



(i) Name this industrial process.

..... [1]

(ii) State the meaning of the symbol $\rightleftharpoons$.

..... [1]

(iii) State the conditions used in this industrial process. Include units.

temperature

pressure

[2]

(iv) Name the catalyst used in this industrial process.

..... [1]

(v) If the pressure is increased, the yield of ammonia increases.

Explain why, in terms of equilibrium.

.....
.....
..... [2]

(vi) If the temperature is increased, the rate of reaction increases.

Explain why, in terms of particles.

.....
.....
.....
.....
..... [3]

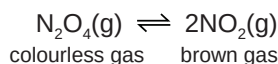
(b) Ammonia reacts with sulfuric acid to make a compound which is used as a fertiliser.

Write the chemical equation for the reaction between ammonia and sulfuric acid.

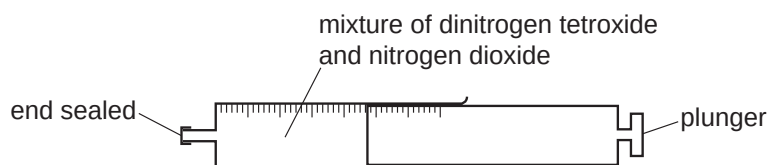
..... [2]

[Total: 12]

- 4 Dinitrogen tetroxide, N_2O_4 , decomposes into nitrogen dioxide, NO_2 . The reaction is reversible.



A gas syringe containing a mixture of dinitrogen tetroxide and nitrogen dioxide gases was sealed and heated. After reaching equilibrium the mixture was a pale brown colour.

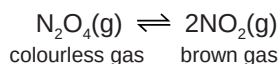


- (a) State what is meant by the term *equilibrium*.

.....

 [2]

- (b) The plunger of the gas syringe is pushed in. The temperature does not change. The mixture initially turns darker brown. After a few seconds the mixture turns lighter brown because the equilibrium shifts to the left.



- (i) Explain why the mixture initially turns darker brown.

..... [1]

- (ii) Explain why the position of equilibrium shifts to the left.

..... [1]

- (c) The forward reaction is endothermic.

- (i) State what happens to the position of equilibrium when the temperature of the mixture is increased.

..... [1]

- (ii) State what happens to the rate of the forward reaction and the rate of the backward reaction when the temperature of the mixture is increased.

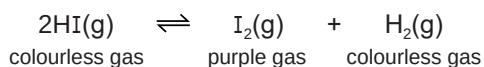
rate of the forward reaction

rate of the backward reaction

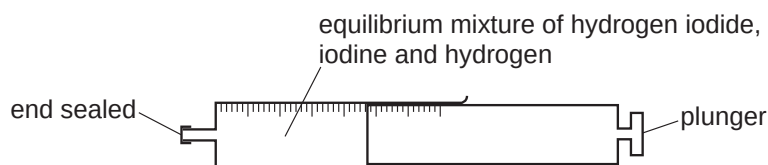
[2]

[Total: 7]

- 4 Hydrogen iodide, HI, decomposes into iodine and hydrogen. The reaction is reversible.



A gas syringe containing a mixture of hydrogen iodide, iodine and hydrogen gases was sealed. After reaching equilibrium the mixture was a pale purple colour.

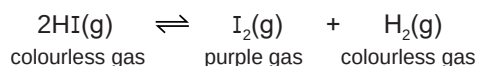


- (a) State what is meant by the term *equilibrium*.

.....
.....
..... [2]

- (b) The plunger of the gas syringe is pushed in. The position of equilibrium does not change. The colour of the gaseous mixture turns darker purple.

The temperature remains constant.



- (i) Explain why the position of equilibrium does **not** change.

..... [1]

- (ii) Suggest why the colour of the gaseous mixture turns darker purple even though the position of equilibrium does not change.

..... [1]

- (c) The forward reaction is endothermic.

- (i) State what happens to the position of equilibrium when the temperature is decreased.

.....
..... [1]

- (ii) State what happens to the rate of the forward reaction and the rate of the backward reaction when the temperature of the mixture is decreased.

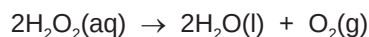
rate of the forward reaction

rate of the backward reaction

[2]

[Total: 7]

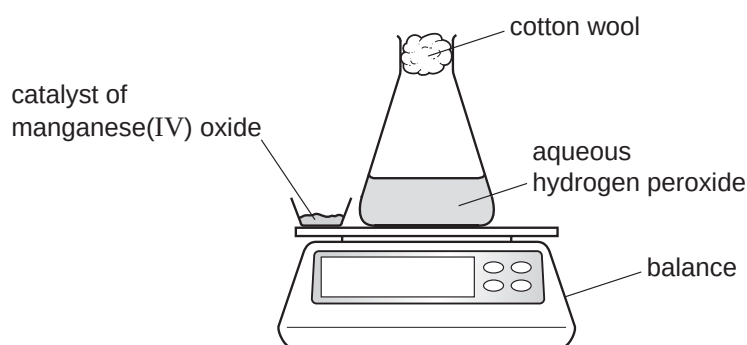
- 6 A student investigates the decomposition of hydrogen peroxide in the presence of a catalyst of manganese(IV) oxide.



- (a) State the meaning of the term *catalyst*.

.....
..... [2]

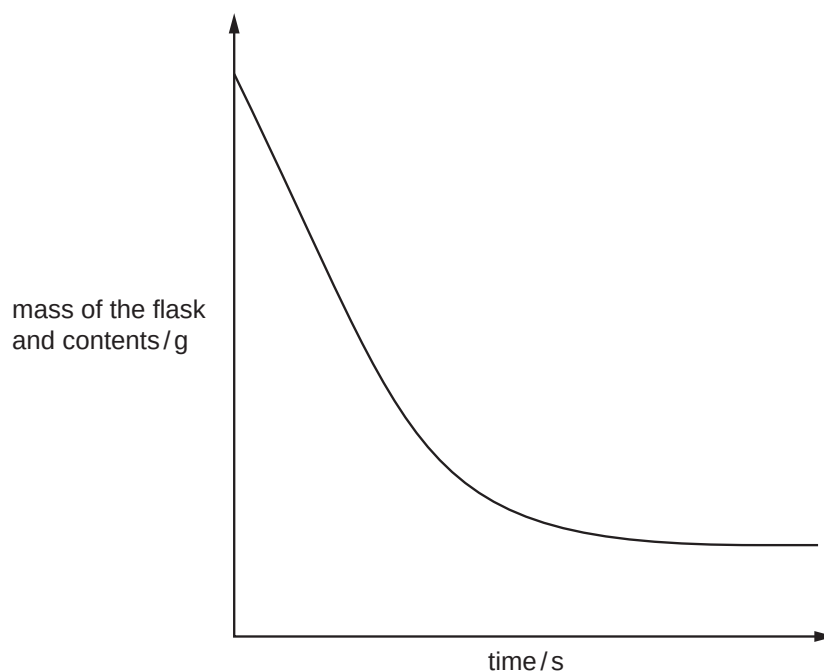
- (b) The diagram shows the equipment the student uses.



The student uses this method:

- the catalyst is added to the aqueous hydrogen peroxide
- the stop-clock is started
- the mass of the flask and contents is recorded at regular time intervals.

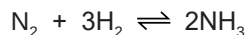
A graph of the mass of the flask and contents against time is shown.



- (i) Suggest why the mass of the flask and contents decreases as time increases.
- [1]
- (ii) Describe what happens to the rate of the reaction as time increases.
-
- [2]
- (c) The student repeats the experiment at a higher temperature. All other conditions stay the same. The rate of reaction increases.
- (i) Explain, in terms of collisions between particles, why the rate of reaction increases at a higher temperature.
-
-
-
- [3]
- (ii) Draw a line on the graph in (b) for the experiment at a higher temperature. [2]

[Total: 10]

- 3 Ammonia is made in an industrial process starting with nitrogen. The equation for the reaction is shown.



- (a) Name the industrial process used to make ammonia.

..... [1]

- (b) State the raw material from which nitrogen is obtained.

..... [1]

- (c) State what is meant by the symbol $\rightleftharpoons$.

..... [1]

- (d) State the temperature and pressure used in this industrial process.

temperature = °C

pressure = atm
[2]

- (e) Name the catalyst used in this industrial process.

..... [1]

- (f) The forward reaction is exothermic.

State the effect, if any, on the position of the equilibrium when the following changes are made.
Explain your answers.

temperature is reduced

.....
.....

pressure is reduced

.....
.....
[4]

- (g) Explain, in terms of particles, what happens to the rate of reaction when the temperature is reduced.

.....
.....
.....
.....
..... [3]

- (h) Give the formula of the compound formed when sulfuric acid reacts with ammonia.

..... [1]

[Total: 14]

- 4 Oxygen is produced by the decomposition of aqueous hydrogen peroxide. Manganese(IV) oxide, MnO_2 , is a catalyst for this reaction.

(a) State the meaning of the term catalyst.

.....
..... [2]

- (b) A student adds powdered manganese(IV) oxide to aqueous hydrogen peroxide in a conical flask as shown in Fig. 4.1. The mass of the conical flask and its contents is measured at regular time intervals. The mass decreases as time increases.

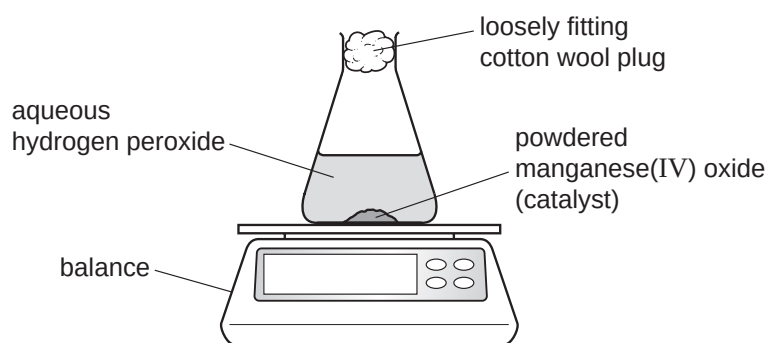


Fig. 4.1

- (i) State why the mass of the conical flask and its contents decreases as time increases.

..... [1]

- (ii) The rate of reaction is highest at the start of the reaction. The rate decreases and eventually becomes zero.

Explain why the rate of reaction is highest at the start of the reaction.

.....
..... [1]

- (iii) Explain why the rate of reaction eventually becomes zero.

.....
..... [1]

- (c) The experiment is repeated at an increased temperature.
All other conditions stay the same.

Explain in terms of collision theory why the rate of reaction is higher at an increased temperature.

.....
.....
.....
..... [3]

- (d) The equation for the decomposition of aqueous hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$, is shown.



50.0 cm³ of a 0.200 mol/dm³ solution of $\text{H}_2\text{O}_2(\text{aq})$ is used.

Calculate the mass of O_2 that forms.
Use the following steps.

- Calculate the number of moles of H_2O_2 used.

..... mol

- Determine the number of moles of O_2 produced.

..... mol

- Calculate the mass of O_2 produced.

..... g
[3]

- (e) State the effect on the mass of oxygen produced if the mass of powdered manganese(IV) oxide catalyst is increased.

..... [1]

- (f) Oxygen can also be produced by the decomposition of mercury(II) oxide, HgO .
The only products of this decomposition are mercury and oxygen.

Write a symbol equation for this decomposition.

..... [2]

[Total: 14]

3 Over 200 million tonnes of sulfuric acid are manufactured every year.

(a) State the name of the process used to manufacture sulfuric acid.

..... [1]

(b) Part of the manufacture of sulfuric acid involves converting sulfur dioxide to sulfur trioxide.

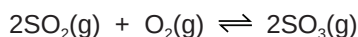
(i) Describe **two** methods by which sulfur dioxide is obtained.

1

2

[2]

The conversion of sulfur dioxide to sulfur trioxide is a reversible reaction which can reach equilibrium.



(ii) State **two** features of an equilibrium.

1

2

[2]

(iii) State the typical conditions and name the catalyst used in the conversion of sulfur dioxide to sulfur trioxide.

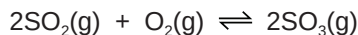
temperature °C

pressure kPa

catalyst

[3]

- (iv) Complete Table 3.1 to show the effect, if any, when the following changes are applied to the conversion of sulfur dioxide to sulfur trioxide.



The forward reaction is exothermic.

Only use the words **increases**, **decreases** or **no change**.

Table 3.1

change	effect on the rate of the forward reaction	effect on the concentration of $\text{SO}_3(\text{g})$ at equilibrium
temperature decreases	decreases	
pressure increases		
no catalyst	decreases	

[4]

- (v) Explain in terms of collision theory why reducing the temperature decreases the rate of the forward reaction.

.....

.....

.....

.....

..... [3]

- (c) Sulfuric acid contains SO_4^{2-} ions.

The oxidation number of O atoms in SO_4^{2-} ions is -2 .

Determine the oxidation number of S atoms in SO_4^{2-} ions. Show your working.

oxidation number = [2]

[Total: 17]

- 5 Methane reacts with steam to produce hydrogen gas.



The reaction takes place at 1000 °C and 100 kPa pressure.

- (a) The reaction is reversible and reaches an equilibrium in a closed system.

State **two** features of an equilibrium.

1

2 [2]

- (b) State and explain, in terms of equilibrium, what happens to the concentration of hydrogen when:

- (i) the pressure is increased

.....
..... [2]

- (ii) the temperature is increased

.....
..... [2]

- (iii) a catalyst is used.

.....
..... [2]

- (c) Methane is a greenhouse gas which contributes to global warming.

- (i) Name a greenhouse gas found in clean, dry air.

..... [1]

- (ii) Explain, in terms of thermal energy, how greenhouse gases cause global warming.

.....
.....
.....
.....
..... [3]

[Total: 12]

- 4 Hydrogen iodide thermally decomposes into iodine and hydrogen. The reaction is reversible.

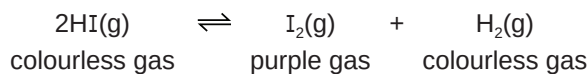


Fig. 4.1 shows a gas syringe containing a mixture of hydrogen iodide, iodine and hydrogen gases. The gas syringe is sealed and the mixture is heated to 300 °C. The mixture of gases reaches equilibrium and is purple.

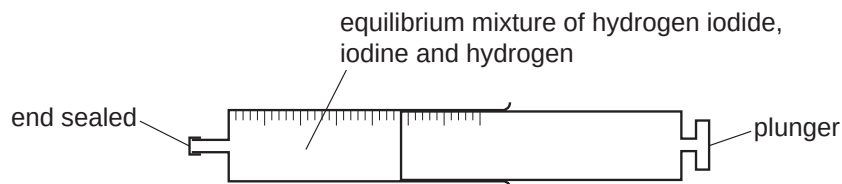


Fig. 4.1

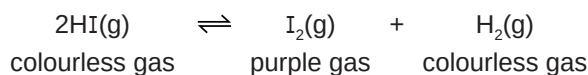
- (a) State what is meant by the term equilibrium.

.....
.....
..... [2]

- (b) The pressure of the mixture is increased. All other conditions stay the same. The position of equilibrium does **not** change.

The colour of the gaseous mixture turns darker purple.

The temperature remains constant.



- (i) Explain why the position of equilibrium does **not** change.

.....
..... [1]

- (ii) Suggest why the colour of the mixture of gases turns darker purple.

.....
..... [1]

- (c) The temperature of the mixture of gases is decreased. All other conditions stay the same.

The mixture of gases turns lighter purple.

State what can be deduced about the forward reaction from this information.

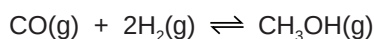
..... [1]

- (d) Deduce the oxidation number of iodine, I, in:

HI

I₂. [2]

- (e) Methanol is manufactured by reacting carbon monoxide with hydrogen.



The rate of formation of methanol increases when a catalyst is used.

- (i) Choose from the list the element that is most likely to be used as the catalyst.

Draw a circle around your chosen answer.

calcium carbon copper sodium sulfur [1]

- (ii) State the effect on the position of equilibrium when a catalyst is used.

..... [1]

- (iii) State the effect that a catalyst has on the activation energy, E_a , of a reaction.

..... [1]

[Total: 10]

- 4 Dinitrogen tetroxide, N_2O_4 , decomposes into nitrogen dioxide, NO_2 . The reaction is reversible.

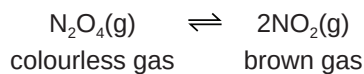


Fig. 4.1 shows a gas syringe containing a mixture of dinitrogen tetroxide and nitrogen dioxide. The gas syringe is sealed. The mixture reaches equilibrium and the colour of the mixture of gases is a pale brown.

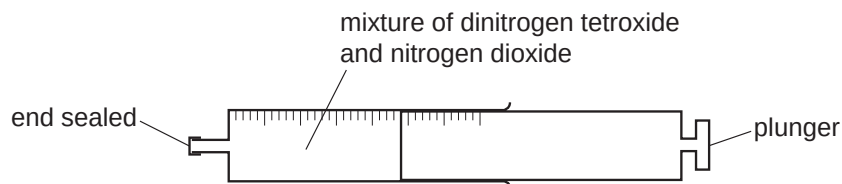


Fig. 4.1

- (a) Describe a reversible reaction at equilibrium in terms of:

- the rate of the forward reaction and the rate of the reverse reaction

.....

- the concentration of reactants and products.

.....

[2]

- (b) The pressure of the mixture is increased. All other conditions stay the same.

The mixture immediately turns darker brown before the position of equilibrium changes.

Explain in terms of particles why the mixture immediately turns darker brown.

..... [1]

- (c) The temperature of the mixture is increased. All other conditions stay the same.

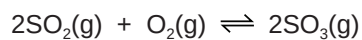
The mixture turns darker brown.

State what can be deduced about the forward reaction from this information.

..... [1]

(d) Sulfur is converted into sulfuric acid, H_2SO_4 , by a series of reactions.

Sulfur dioxide, SO_2 , and oxygen, O_2 , react to form sulfur trioxide, SO_3 . The reversible reaction reaches equilibrium.



(i) Complete Table 4.1 using only the words, **increases**, **decreases** or **no change**.

Table 4.1

	effect on the rate of the forward reaction	effect on the equilibrium yield of $\text{SO}_3(\text{g})$
add a catalyst		
increase the pressure		

[4]

(ii) Deduce the oxidation number of sulfur in:

S

SO_3

[2]

[Total: 10]

- 3 Sulfur forms two chlorides, **P** and **Q**.
Chloride **P** has the formula S_2Cl_2 . Chloride **Q** has the formula SCl_2 .

- (a) Both chlorides are covalently bonded and have low melting points.

Suggest, in terms of attraction between particles, why these chlorides have low melting points.

.....
..... [2]

- (b) Chloride **P**, S_2Cl_2 , forms when sulfur reacts with chlorine.

Write the symbol equation for this reaction.

..... [1]

- (c) Complete the dot-and-cross diagram in Fig. 3.1 of a molecule of chloride **Q**, SCl_2 .

Show outer electrons only.

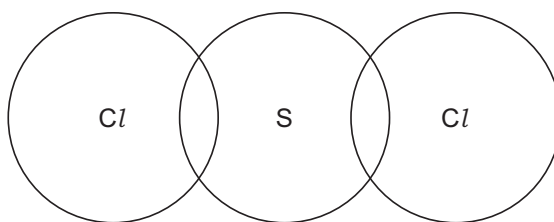
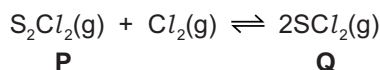


Fig. 3.1

[3]

- (d) Chloride **P** is converted to chloride **Q** by reaction with chlorine in a closed system.
The reversible reaction reaches an equilibrium.



The forward reaction is exothermic.

Suggest **two** changes to the conditions which will result in a decrease in the concentration of chloride **Q** at equilibrium.

1
2 [2]



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(e) The rate of the forward reaction in (d) is determined by collision theory.

The rate of reaction depends upon two factors:

- the frequency of collisions between particles
- the proportion of collisions which have energy greater than or equal to the activation energy.

(i) Define the term activation energy.

..... [1]

(ii) Give the symbol for activation energy.

..... [1]

(iii) Complete Table 3.1 to show the effect, if any, when the conditions are changed.

Use only the words **increases**, **decreases** or **no change**.

Table 3.1

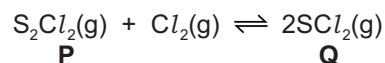
change to conditions	effect on the frequency of collisions between particles	effect on the proportion of collisions which have energy greater than or equal to the activation energy
concentration of chlorine is increased		
temperature is increased		
a catalyst is added		

[5]





(f) The reaction of chloride **P** with chlorine is a redox reaction.



The oxidation number of **Cl** in chloride **P** and chloride **Q** is -1 .

Use oxidation numbers to explain why:

- sulfur is oxidised in the forward reaction

.....
.....

- chlorine is oxidised in the reverse reaction.

.....
.....

[4]

[Total: 19]

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Question 4 starts on the next page.

